JavaScript is disabled on your browser.

compact1, compact2, compact3

java.nio

Class IntBuffer

java.lang.Object
    java.nio.Buffer
        java.nio.IntBuffer

All Implemented Interfaces:

Comparable<IntBuffer>

```
public abstract class IntBuffer
extends Buffer
implements Comparable<IntBuffer>
```

An int buffer.

This class defines four categories of operations upon int buffers:

- Absolute and relative *get* and *put* methods that read and write single ints;
- Relative *bulk get* methods that transfer contiguous sequences of ints from this buffer into an array; and
- Relative *bulk put* methods that transfer contiguous sequences of ints from an int array or some other int buffer into this buffer; and
- Methods for compacting, duplicating, and slicing an int buffer.

Int buffers can be created either by *allocation*, which allocates space for the buffer's content, by *wrapping* an existing int array into a buffer, or by creating a *view* of an existing byte buffer.

Like a byte buffer, an int buffer is either *direct* or *non-direct*. A int buffer created via the wrap methods of this class will be non-direct. An int buffer created as a view of a byte buffer will be direct if, and only if, the byte buffer itself is direct. Whether or not an int buffer is direct may be determined by invoking the isDirect method.

Methods in this class that do not otherwise have a value to return are specified to return the buffer upon which they are invoked. This allows method invocations to be chained.

Since:

1.4

Method Summary

Instance Methods | Abstract Methods | Concrete Methods

Modifier and Type	Method	Description
static IntBuffer	allocate(int capacity)	Allocates a new int buffer.
int[]	array()	Returns the int array that backs this buffer *(optional operation)*.
int	arrayOffset()	Returns the offset within this buffer's backing array of the first element of the buffer *(optional operation)*.
abstract IntBuffer	asReadOnlyBuffer()	Creates a new, read-only int buffer that shares this buffer's content.
abstract IntBuffer	compact()	Compacts this buffer *(optional operation)*.
int	compareTo(IntBuffer that)	Compares this buffer to another.
abstract IntBuffer	duplicate()	Creates a new int buffer that shares this buffer's content.

boolean	**equals**(**Object** ob)	Tells whether or not this buffer is equal to another object.
abstract int	**get**()	Relative *get* method.
abstract int	**get**(int index)	Absolute *get* method.
IntBuffer	**get**(int[] dst)	Relative bulk *get* method.
IntBuffer	**get**(int[] dst, int offset, int length)	Relative bulk *get* method.
boolean	**hasArray**()	Tells whether or not this buffer is backed by an accessible int array.
int	**hashCode**()	Returns the current hash code of this buffer.
abstract boolean	**isDirect**()	Tells whether or not this int buffer is direct.
abstract **ByteOrder**	**order**()	Retrieves this buffer's byte order.
abstract **IntBuffer**	**put**(int i)	Relative *put* method *(optional operation)*.
IntBuffer	**put**(int[] src)	Relative bulk *put* method *(optional operation)*.
IntBuffer	**put**(int[] src, int offset, int length)	Relative bulk *put* method *(optional operation)*.
IntBuffer	**put**(**IntBuffer** src)	Relative bulk *put* method *(optional operation)*.
abstract **IntBuffer**	**put**(int index, int i)	Absolute *put* method *(optional operation)*.

abstract **IntBuffer**	**slice**()	Creates a new int buffer whose content is a shared subsequence of this buffer's content.
String	**toString**()	Returns a string summarizing the state of this buffer.
static **IntBuffer**	**wrap**(int[] array)	Wraps an int array into a buffer.
static **IntBuffer**	**wrap**(int[] array, int offset, int length)	Wraps an int array into a buffer.

Methods inherited from class java.nio.Buffer

capacity, clear, flip, hasRemaining, isReadOnly, limit, limit, mark, position, position, remaining, reset, rewind

Methods inherited from class java.lang.Object

clone, finalize, getClass, notify, notifyAll, wait, wait, wait

Method Detail

allocate

```
public static IntBuffer allocate(int capacity)
```

Allocates a new int buffer.

The new buffer's position will be zero, its limit will be its capacity, its mark will be undefined, and each of its elements will be initialized to zero. It will have a `backing array`, and its `array offset` will be zero.

Parameters:

`capacity - The new buffer's capacity, in ints`

Returns:

`The new int buffer`

Throws:

`IllegalArgumentException - If the capacity is a negative integer`

wrap

```
public static IntBuffer wrap(int[] array,
                             int offset,
                             int length)
```

Wraps an int array into a buffer.

The new buffer will be backed by the given int array; that is, modifications to the buffer will cause the array to be modified and vice versa. The new buffer's capacity will be `array.length`, its position will be `offset`, its limit will be `offset + length`, and its mark will be undefined. Its `backing array` will be the given array, and its `array offset` will be zero.

Parameters:

`array - The array that will back the new buffer`

`offset - The offset of the subarray to be used; must be non-negative and no larger than array.length. The new buffer's position will be set to this value.`

`length - The length of the subarray to be used; must be non-negative and no larger than array.length - offset. The new buffer's limit will be set to offset + length.`

Returns:

`The new int buffer`

Throws:

`IndexOutOfBoundsException - If the preconditions on the offset and length parameters do not hold`

wrap

```
public static IntBuffer wrap(int[] array)
```

Wraps an int array into a buffer.

The new buffer will be backed by the given int array; that is, modifications to the buffer will cause the array to be modified and vice versa. The new buffer's capacity and limit will be `array.length`, its position will be zero, and its mark will be undefined. Its `backing array` will be the given array, and its `array offset>` will be zero.

Parameters:

`array - The array that will back this buffer`

Returns:

`The new int buffer`

slice

```
public abstract IntBuffer slice()
```

Creates a new int buffer whose content is a shared subsequence of this buffer's content.

The content of the new buffer will start at this buffer's current position. Changes to this buffer's content will be visible in the new buffer, and vice versa; the two buffers' position, limit, and mark values will be independent.

The new buffer's position will be zero, its capacity and its limit will be the number of ints remaining in this buffer, and its mark will be undefined. The new buffer will be direct if, and only if, this buffer is direct, and it will be read-only if, and only if, this buffer is read-only.

Returns:

`The new int buffer`

duplicate

```
public abstract IntBuffer duplicate()
```

Creates a new int buffer that shares this buffer's content.

The content of the new buffer will be that of this buffer. Changes to this buffer's content will be visible in the new buffer, and vice versa; the two buffers' position, limit, and mark values will be independent.

The new buffer's capacity, limit, position, and mark values will be identical to those of this buffer. The new buffer will be direct if, and only if, this buffer is direct, and it will be read-only if, and only if, this buffer is read-only.

Returns:

`The new int buffer`

asReadOnlyBuffer

```
public abstract IntBuffer asReadOnlyBuffer()
```

Creates a new, read-only int buffer that shares this buffer's content.

The content of the new buffer will be that of this buffer. Changes to this buffer's content will be visible in the new buffer; the new buffer itself, however, will be read-only and will not allow the shared content to be modified. The two buffers' position, limit, and mark values will be independent.

The new buffer's capacity, limit, position, and mark values will be identical to those of this buffer.

If this buffer is itself read-only then this method behaves in exactly the same way as the `duplicate` method.

Returns:

`The new, read-only int buffer`

get

```
public abstract int get()
```

Relative *get* method. Reads the int at this buffer's current position, and then increments the position.

Returns:

`The int at the buffer's current position`

Throws:

`BufferUnderflowException - If the buffer's current position is not smaller than its limit`

put

```
public abstract IntBuffer put(int i)
```

Relative *put* method *(optional operation)*.

Writes the given int into this buffer at the current position, and then increments the position.

Parameters:

`i - The int to be written`

Returns:

`This buffer`

Throws:

`BufferOverflowException - If this buffer's current position is not smaller than its limit`

`ReadOnlyBufferException - If this buffer is read-only`

get

```
public abstract int get(int index)
```

Absolute *get* method. Reads the int at the given index.

Parameters:

`index - The index from which the int will be read`

Returns:

`The int at the given index`

Throws:

`IndexOutOfBoundsException - If index is negative or not smaller than the buffer's limit`

put

```
public abstract IntBuffer put(int index,
                              int i)
```

Absolute *put* method *(optional operation)*.

Writes the given int into this buffer at the given index.

Parameters:

`index - The index at which the int will be written`

`i - The int value to be written`

Returns:

`This buffer`

Throws:

`IndexOutOfBoundsException - If index is negative or not smaller than the buffer's limit`

`ReadOnlyBufferException - If this buffer is read-only`

get

```
public IntBuffer get(int[] dst,
                     int offset,
                     int length)
```

Relative bulk *get* method.

This method transfers ints from this buffer into the given destination array. If there are fewer ints remaining in the buffer than are required to satisfy the request, that is, if `length > remaining()`, then no ints are transferred and a `BufferUnderflowException` is thrown.

Otherwise, this method copies `length` ints from this buffer into the given array, starting at the current position of this buffer and at the given offset in the array. The position of this buffer is then incremented by `length`.

In other words, an invocation of this method of the form `src.get(dst, off, len)` has exactly the same effect as the loop

```
for (int i = off; i < off + len; i++)
    dst[i] = src.get():
```

except that it first checks that there are sufficient ints in this buffer and it is potentially much more efficient.

Parameters:

`dst - The array into which ints are to be written`

`offset - The offset within the array of the first int to be written; must be non-negative and no larger than dst.length`

`length - The maximum number of ints to be written to the given array; must be non-negative and no larger than dst.length - offset`

Returns:

`This buffer`

Throws:

`BufferUnderflowException - If there are fewer than length ints remaining in this buffer`

`IndexOutOfBoundsException - If the preconditions on the offset and length parameters do not hold`

get

`public IntBuffer get(int[] dst)`

Relative bulk *get* method.

This method transfers ints from this buffer into the given destination array. An invocation of this method of the form `src.get(a)` behaves in exactly the same way as the invocation

```
src.get(a, 0, a.length)
```

Parameters:

`dst - The destination array`

Returns:

`This buffer`

Throws:

`BufferUnderflowException - If there are fewer than length ints remaining in this buffer`

put

```
public IntBuffer put(IntBuffer src)
```

Relative bulk *put* method *(optional operation)*.

This method transfers the ints remaining in the given source buffer into this buffer. If there are more ints remaining in the source buffer than in this buffer, that is, if `src.remaining() > remaining()`, then no ints are transferred and a `BufferOverflowException` is thrown.

Otherwise, this method copies *n* = `src.remaining()` ints from the given buffer into this buffer, starting at each buffer's current position. The positions of both buffers are then incremented by *n*.

In other words, an invocation of this method of the form `dst.put(src)` has exactly the same effect as the loop

```
    while (src.hasRemaining())
        dst.put(src.get());
```

except that it first checks that there is sufficient space in this buffer and it is potentially much more efficient.

Parameters:

`src - The source buffer from which ints are to be read; must not be this buffer`

Returns:

`This buffer`

Throws:

`BufferOverflowException - If there is insufficient space in this buffer for the remaining ints in the source buffer`

`IllegalArgumentException - If the source buffer is this buffer`

`ReadOnlyBufferException - If this buffer is read-only`

put

```
public IntBuffer put(int[] src,
                     int offset,
                     int length)
```

Relative bulk *put* method *(optional operation)*.

This method transfers ints into this buffer from the given source array. If there are more ints to be copied from the array than remain in this buffer, that is, if `length > remaining()`, then no ints are transferred and a `BufferOverflowException` is thrown.

Otherwise, this method copies `length` ints from the given array into this buffer, starting at the given offset in the array and at the current position of this buffer. The position of this buffer is then incremented by `length`.

In other words, an invocation of this method of the form `dst.put(src, off, len)` has exactly the same effect as the loop

```
for (int i = off; i < off + len; i++)
    dst.put(a[i]);
```

except that it first checks that there is sufficient space in this buffer and it is potentially much more efficient.

Parameters:

`src - The array from which ints are to be read`

`offset - The offset within the array of the first int to be read; must be non-negative and no larger than array.length`

`length - The number of ints to be read from the given array; must be non-negative and no larger than array.length - offset`

Returns:

`This buffer`

Throws:

`BufferOverflowException - If there is insufficient space in this buffer`

`IndexOutOfBoundsException - If the preconditions on the offset and length parameters do not hold`

`ReadOnlyBufferException - If this buffer is read-only`

put

`public final IntBuffer put(int[] src)`

Relative bulk *put* method *(optional operation)*.

This method transfers the entire content of the given source int array into this buffer. An invocation of this method of the form `dst.put(a)` behaves in exactly the same way as the invocation

```
dst.put(a, 0, a.length)
```

Parameters:

`src - The source array`

Returns:

`This buffer`

Throws:

`BufferOverflowException` - If there is insufficient space in this buffer

`ReadOnlyBufferException` - If this buffer is read-only

hasArray

```
public final boolean hasArray()
```

Tells whether or not this buffer is backed by an accessible int array.

If this method returns `true` then the `array` and `arrayOffset` methods may safely be invoked.

Specified by:

`hasArray` in class `Buffer`

Returns:

`true if, and only if, this buffer is backed by an array and is not read-only`

array

```
public final int[] array()
```

Returns the int array that backs this buffer *(optional operation)*.

Modifications to this buffer's content will cause the returned array's content to be modified, and vice versa.

Invoke the `hasArray` method before invoking this method in order to ensure that this buffer has an accessible backing array.

Specified by:

`array` in class `Buffer`

Returns:

`The array that backs this buffer`

Throws:

`ReadOnlyBufferException` - If this buffer is backed by an array but is read-only

`UnsupportedOperationException` - If this buffer is not backed by an accessible array

arrayOffset

```
public final int arrayOffset()
```

Returns the offset within this buffer's backing array of the first element of the buffer *(optional operation)*.

If this buffer is backed by an array then buffer position *p* corresponds to array index *p* + `arrayOffset()`.

Invoke the `hasArray` method before invoking this method in order to ensure that this buffer has an accessible backing array.

Specified by:

`arrayOffset` in class `Buffer`

Returns:

The offset within this buffer's array of the first element of the buffer

Throws:

`ReadOnlyBufferException` - If this buffer is backed by an array but is read-only

`UnsupportedOperationException` - If this buffer is not backed by an accessible array

compact

```
public abstract IntBuffer compact()
```

Compacts this buffer *(optional operation)*.

The ints between the buffer's current position and its limit, if any, are copied to the beginning of the buffer. That is, the int at index *p* = `position()` is copied to index zero, the int at index *p* + 1 is copied to index one, and so forth until the int at index `limit()` - 1 is copied to index *n* = `limit()` - 1 - *p*. The buffer's position is then set to *n+1* and its limit is set to its capacity. The mark, if defined, is discarded.

The buffer's position is set to the number of ints copied, rather than to zero, so that an invocation of this method can be followed immediately by an invocation of another relative *put* method.

Returns:

This buffer

Throws:

`ReadOnlyBufferException` - If this buffer is read-only

isDirect

```
public abstract boolean isDirect()
```

Tells whether or not this int buffer is direct.

Specified by:

`isDirect` in class `Buffer`

Returns:

`true if, and only if, this buffer is direct`

toString

```
public String toString()
```

Returns a string summarizing the state of this buffer.

Overrides:

`toString` in class `Object`

Returns:

`A summary string`

hashCode

```
public int hashCode()
```

Returns the current hash code of this buffer.

The hash code of a int buffer depends only upon its remaining elements; that is, upon the elements from `position()` up to, and including, the element at `limit() - 1`.

Because buffer hash codes are content-dependent, it is inadvisable to use buffers as keys in hash maps or similar data structures unless it is known that their contents will not change.

Overrides:

`hashCode` in class `Object`

Returns:

`The current hash code of this buffer`

See Also:

`Object.equals(java.lang.Object)`, `System.identityHashCode(java.lang.Object)`

equals

```
public boolean equals(Object ob)
```

Tells whether or not this buffer is equal to another object.

Two int buffers are equal if, and only if,

1. They have the same element type,
2. They have the same number of remaining elements, and
3. The two sequences of remaining elements, considered independently of their starting positions, are pointwise equal.

A int buffer is not equal to any other type of object.

Overrides:

`equals` in class `Object`

Parameters:

`ob - The object to which this buffer is to be compared`

Returns:

`true if, and only if, this buffer is equal to the given object`

See Also:

`Object.hashCode()`, `HashMap`

compareTo

```
public int compareTo(IntBuffer that)
```

Compares this buffer to another.

Two int buffers are compared by comparing their sequences of remaining elements lexicographically, without regard to the starting position of each sequence within its corresponding buffer. Pairs of `int` elements are compared as if by invoking `Integer.compare(int,int)`.

A int buffer is not comparable to any other type of object.

Specified by:

`compareTo` in interface `Comparable<IntBuffer>`

Parameters:

`that` - the object to be compared.

Returns:

A negative integer, zero, or a positive integer as this buffer is less than, equal to, or greater than the given buffer

order

```
public abstract ByteOrder order()
```

Retrieves this buffer's byte order.

The byte order of an int buffer created by allocation or by wrapping an existing `int` array is the `native order` of the underlying hardware. The byte order of an int buffer created as a view of a byte buffer is that of the byte buffer at the moment that the view is created.

Returns:

```
This buffer's byte order
```


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